

## Response to the Reviewers

**Manuscript** 1568909 — *A 3D Multiphase Continuum Model for Atoms* **Revised title** *A 3D Multiphase Continuum Model for Atoms and the Periodic Table* **Journal** International Journal of Quantum Chemistry **Decision** Major revision

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We thank both reviewers. On the points that mattered most they were correct, and we have not argued with either: Reviewer 2's objection to the packing argument and Reviewer 1's "elephant in the room" both identified places where the framing claimed more than the body text supported.

Two things about the revision should be said before the point-by-point replies, and we keep them short here because both are presented properly in the manuscript.

**A new account of the build-up of the periodic table** (§4.3). The table is assembled from two packings the calculations themselves produce — the antipodal pair, 2, and the octet as two dual tetrahedra,  $8 = 4+4$  — and from one constraint: a row ending in 4 or 6 would need half an octet or three quarters of one, which is not a packing available to the model, so the remainder after octets is forced to 0 or 2. That gives every row length 2, 8, 8, 18, 18, 32, 32 and every closed shell 2, 10, 18, 36, 54, 86, 118, with  $2n^2$  nowhere in the construction — so its agreement with the hydrogenic count is a result, not an input. Three consequences are new: the octet is the unit every later row is built from (18 a pair and two octets, 32 four octets); the step before closure is  $4+3$ , so a halogen is short of a tetrahedral *site* rather than of a count; and  $\sqrt{\text{cap}(n+1)/\text{cap}(n)} = (n+1)/n$  holds exactly when each capacity serves two periods, which is a direction on the period doubling and is labelled as no more than that. We claim a decomposition of the capacities, not a derivation of the row lengths; that each capacity serves two periods remains the principal open item. The  $4+4$  octet is Linnett's double quartet [*J. Am. Chem. Soc.* **83** (1961) 2643], which separates the quartets by spin; this model has none and reaches the same arrangement from Coulomb repulsion and non-overlap alone.

**Periods 1–4, with no parameters at all.** The energies run hydrogen to krypton,  $Z = 1-36$ , all-electron with the bare term  $-Z/r$ : no core radius, no screened kernel, no frozen core, no fitted length, nothing calibrated against a reference atom, nothing set per element. The frozen core and pseudo-kernel of the original submission, with the softening radius  $r_c$  Reviewer 2 identified, are gone; resolution recovers the range in their place. The only quantity that varies across the table is the mesh —  $N = 400$  through argon,  $N = 800$  from potassium, since  $r_{1s} \approx 4.3/Z$  and no single mesh serves both ends — and a mesh is a discretisation, not a parameter: it carries no physics and is fitted to nothing.

**A word on length.** The manuscript has grown from 13 pages to 35, and we should say plainly why, since a revision asked for retractions does not usually come back close to three times the size. The reason is not that the original argument has been padded. It is that the paper now makes a different kind of claim: the original computed energies for atoms, and this version offers an account of the periodic table — every row length and every closure derived from two filled packings, the antipodal pair and the  $4+4$  octet, with no  $2n^2$  entering anywhere.

A claim of that kind cannot be made briefly, because it is checkable only if it is complete. It has to produce every capacity and every closure rather than the convenient ones; it has to meet the cases that do not come out as one would wish, and krypton is one — its outermost eight is not  $4+4$  under either filling rule we use, nor in the orbital account, and we say so and show why closure is exhaustion of a row's capacity rather than the shape of its outer layer. It has to separate what is derived from what is supplied, state exactly where the build-up stops, and report the measurements that bear on the open items together with what they fail to establish. A shorter version would be a weaker claim, not the same claim told faster, and the referees' objections were in substance that the original claimed more than it had laid out.

It should be said where the construction came from, because it was not an addition we chose to

make. It is the answer to Reviewer 2's second question. Asked why cap-4 configurations are used when the radial model favours cap-8, we found we could not answer in the terms the original paper had: that cap-4 agrees better with reference energies is a reason to prefer a number, not a reason for the number to be what it is. Answering properly meant saying what makes eight a closed shell at all — and that is the inversion argument. A single tetrahedron has  $T_d$  symmetry and no centre of inversion, so it is directional and cannot close; its dual is a distinct arrangement, and the union of the two is the cube, which is centrosymmetric; the sequence stops there, the cube being its own inversion. So 4 cannot close and 8 can. The pair and the octet follow from that, the capacities follow from the pair and the octet, and the row lengths and closures follow from the capacities. The account of the periodic table in this version exists because that question was put to us, and the same is true of most of what else is new: the all-electron solves answer R2.1, the radial-nesting section answers R2.4, and the Aufbau measurements answer the half of R2.2 that asks how a configuration is selected at all.

The material we withdrew is substantial — both heavy-element tables, all the ionization-energy material, the frozen core and the pseudo-kernel — and the growth is entirely in what replaced it: the pair-and-octet construction, the Aufbau section with its measurements, three figures of the build-up in three dimensions, the fourth-row table, electronegativity as the valence observable, and the analysis of what the spherical reduction costs. None of it was in the original, and most of it exists because the reviewers' objections pointed there. **If the editor considers the paper too long, we are willing to shorten it** and would propose cutting the discussion of packing versus orbitals and the more expository passages, leaving the construction, the tables and the figures intact.

## Reviewer 1

**R1.1 — The ontological status of RealQM. The justification is deferred to a reference that is itself under review, and the claim that RealQM is a correction to rather than an approximation of standard quantum mechanics cannot rest on atomic results alone. At minimum the ontological claims should be phrased as speculation.**

We accept this without reservation and have taken the off-ramp the reviewer offered. Section 1 now closes with a paragraph (*Comparison with textbook quantum mechanics*) stating that the two are different mathematical objects — a linear Schrödinger equation for one wave function on  $3N$ -dimensional configuration space, versus a nonlinear free-boundary problem for  $N$  charge densities on three-dimensional physical space — that exclusion is realised differently in each, and that **neither is shown here to be an approximation to the other**. It states explicitly that the wider ontological reading advanced elsewhere should be taken as speculative pending evidence stronger than atomic energetics can supply, and that nothing in this paper depends on it. What is compared here is what each produces for the periodic table, and nothing more.

**R1.2 — Could equivalence with density functional theory be established?**

Not attempted, and we should say why we do not expect it to be a short step. RealQM is fundamentally different from DFT in the object it works with. DFT is built on a single common density  $\rho = \sum_i |\psi_i|^2$ , and Hohenberg-Kohn is a statement about that one function. RealQM has no such object in that role: its state is  $N$  *individual* charge densities together with the partition of space between them, which carries strictly more information than their sum. Two different partitions of the same total density have different energies, because the  $i \neq j$  rule counts pairs over disjoint domains rather than integrating a common  $\rho$  against itself.

The consequence is visible. DFT's Hartree term integrates  $\rho$  against itself and so carries a spurious self-interaction, which the exchange-correlation functional is partly there to remove; here there is nothing to remove. An equivalence would have to relate objects of different kinds, and

relate an approximated functional to one written down exactly. We name it as open.

**R1.3 — Does the method scale to molecules? Is the spherical model useful beyond atoms?**

The spherical reduction is specific to atoms and does not extend; it is available here only because an atom is spherically symmetric, and the revised introduction now derives the reduction from that fact rather than presenting it as an implementation choice. The three-dimensional form of the model carries no such restriction and is the one that extends.

On molecules, what the method supplies first is **forces**. The free boundaries move with the charge, so the force on each kernel is available at every step of the same relaxation, with no separate apparatus and no derivative of a fitted functional; geometry follows from it, and molecular dynamics follows without further machinery. Energy differences within one composition are reliable for the same reason, the discretisation error being common to the states compared. Absolute energetics across compositions is the harder case and we make no claim about it here. No molecular result is reported in this paper.

**R1.4 — Could basis sets reduce the cost, particularly for periodic systems?**

We do not think a basis is needed; the mesh is enough. A basis exists to make the integrals of a configuration-space method tractable and to compress the representation, and neither problem arises: there are no two-electron integrals, the interaction coming from a Poisson solve on the same grid, and the cost is already linear in the number of electrons.

There is also a reason of principle. The unknowns include the interfaces, which are geometric: a domain boundary is a moving surface, represented directly by a mesh, where a global basis would have to build a nearly discontinuous partition out of smooth functions. And a basis is an empirical input whose adequacy must be established per system; introducing one would give back exactly the parameter-free character R2.1 asks us to secure. Where more accuracy is wanted we refine the mesh.

**R1.5 — Is the imaginary-time propagation connected to path-integral or centroid formalisms?**

No. It is a relaxation scheme used to reach the stationary state and has no such connection. Only the converged solution enters any result, so the temporal ordering of the relaxation is immaterial. This is now stated in §Limitations.

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## Reviewer 2

**R2.1 — The abstract claims an in-principle parameter-free calculation, but the results use per-period  $r_c$  calibrations, and separate ones for total energies and ionization energies.**

Correct, and this was the decisive report on the paper. The two statements in the original manuscript could not both stand. As described above, we did not repair the claim — we removed the construction that violated it. There is no  $r_c$  anywhere in the revised paper, no calibration of any kind, and no per-element or per-period empirical input. The inputs are the nuclear charge and the number of electrons.

What made that possible is worth reporting, because it was not obvious to us. The calibrated radius existed to let the heavier elements be computed on a fixed mesh; its real function was to remove the innermost shell, which is the one a uniform grid resolves worst. But that is a resolution problem and it has a resolution answer. The free-boundary shell radii are outputs of the model, and we now measure the innermost one to follow  $r_{1s}Z \approx 4.3$ , constant to  $\pm 9\%$  from carbon to argon. Refining the mesh accordingly — and scaling the step count with it, since the relaxation is

explicit — the all-electron energies run from argon to zinc within 1–5% with nothing fitted. The parameter was standing in for a mesh.

(The original abstract also asserted a core radius scaling as  $1/Z$  as a *result*. The body never supported it and it has been withdrawn; what is stated above is a measurement used to choose a grid, not a claim about the model’s output.)

**R2.2 — The radial model favours cap-8 packing, yet cap-4 configurations are chosen because they agree better with reference energies.**

The reviewer’s reading was accurate: the packing was selected by fit. The revision replaces that selection with two things — a measurement of why the energy cannot decide, and a construction of the capacities that does not appeal to energies at all.

**The energy is monotone in the packing, and anti-correlated with the physical configuration.** There is no interior minimum, so an unconstrained minimiser absorbs the core into the valence. Beryllium’s three candidates order by compactness: a single 4-shell at  $-18.893$  Ha (28.8% too deep), 2+2 at  $-14.893$  (1.54% too deep), 2+1+1 at  $-14.603$  (0.44% short). The lower energy is reached by over-binding past the reference, so the physical configuration is the least bound and the closest — and an energy preference could not have supported it in any case. The 2+2 margin is converged, on a step ladder flat from 16,000 to 64,000 steps; it is the monotonicity, not a search failure.

**What decides instead is the relaxation.** The algorithm does not minimise over configurations; it reaches the equilibrium at the end of its path. A filled shell is more rigid than the charge arriving at it, and the interface moves only where one domain’s amplitude locally exceeds the other’s, so the newcomer is turned aside by a comparison of amplitudes and not by an imposed rule. Figures 2 and 3 show this in three dimensions, all-electron, with no  $r_c$ : a charge launched as a half-shell wraps a  $\text{Li}^+$  core and closes into a shell, declining a merged state that was reachable throughout and lies 1.055 Ha *lower*. A seed ladder rules out the obvious objection that the interface merely stayed where it was put — seeds 60% apart give shell radii 2.139 and 2.153 a.u., 0.7% apart.

**The capacities are not fitted either**, and this is the part we would now put forward as the paper’s contribution. They are built from the two filled packings the calculations establish — the antipodal pair and the dual-tetrahedral octet, the latter against a single 8-shell that over-binds by 9–22% — together with the constraint that a row cannot end in 4 or 6. Every row length and every closed shell follows, without  $2n^2$  entering. §4.3 gives the construction, the 4+3 step before closure, the statement that only a noble gas can carry a complete outermost octet, and the direction on the doubling. One classical result is conceded there rather than left for a reader to catch: for eight points on *one* sphere the Coulomb minimum is the square antiprism, not the cube, by 0.33%, and that is a theorem — though the antiprism is itself a 4+4, and this model does not place eight charges on one sphere.

What we do *not* claim is that the row lengths are derived. The capacities decompose into pairs and octets uniquely, and that is established; that the third row has length 8 and the fourth 18 is not.

**R2.3 — Equation (1) identifies  $-\varepsilon_i$  with the optimised neutral-ion energy difference. Why should these be equal?**

They are not, and we measured the discrepancy before deciding what to do about it. For bare-nucleus balanced configurations, Koopmans  $-\varepsilon_i$  against  $\Delta\text{SCF}$  against NIST (eV):

|   | Koopmans | $\Delta\text{SCF}$ | NIST  |
|---|----------|--------------------|-------|
| C | 4.85     | 8.74               | 11.26 |
| O | 6.79     | 12.04              | 13.62 |

The frozen estimate is roughly half the relaxed one, so the identification in the original Eq. (1) was unsupported. Worse for the method,  $\Delta\text{SCF}$  cannot simply replace it: it requires choosing the

cation’s packing, and for nitrogen neither choice works — repacking gives +51.4 eV, decrementing the outer shell gives a state *below* the neutral and a negative ionization energy.  **$\Delta$ SCF therefore inherits the packing-selection problem of R2.2.**

Given this, we concluded that ionization is the wrong observable for a ground-state model: it is a *process*, in which the remaining electrons repack, and the model has no principled way to select the ion’s packing. **All ionization-energy material has been removed from the paper**, and the abstract and Scope now state that ionization and excitation are left aside, with that reason given.

In its place we report a quantity the model is built to answer: the **Allen configuration energy**  $\chi = (n_s \varepsilon_s + n_p \varepsilon_p) / (n_s + n_p)$ , computed natively from the shell energies. It is a state property of the neutral atom — no ion, no anion, no relaxation, no calibration — and it is what bond polarity is built from. The **ordering across period 2 is exact**. The gradient is too steep by a factor 2.23 (span 51.1 against 22.9 eV), with the error concentrated at the electronegative end and climbing monotonically from 1.09 at Be to 2.06 at Ne. We present the result where it is strong and state the rest as the known light-*p*-block limitation, whose structural cause (the outer tetrahedron placed at too large a radius in a radial model) and cure (the three-dimensional solve) are both identified.

**R2.4 — How were the transition-metal points of Figure 1 obtained, and do they establish the claimed 4s-before-3d removal order?**

They do not. On checking the generating code we found the  $s^2$  shell is placed **outermost by construction**; the run then tests whether the outermost shell is the *s* shell, which in a nested radial model the input very nearly guarantees. The reviewer’s suspicion was correct: 4s-before-3d was an input, not a result. Two further defects: those runs used 4000 relaxation steps, which is below this project’s own convergence floor, and their total energies were floor-carried and not independently meaningful.

**The claim has been removed from the abstract and from the paper.** The *d*-block is now outside the stated scope entirely, and §*Boundary of the spherical model* reports the 4s/3d interpenetration as the structure the radial reduction cannot represent — described as a limitation, with no ordering claimed as a result. §Limitations states that the placement of the *d*-shell inside the *s*-shell is imposed, not derived.

**R2.5 — The Kr→Xe increase contradicts the claimed down-group trend.**

**R2.6 — Argon appears as 11.29 eV in Table 3 and 8.0 eV in Table 5.**

**R2.7 — *p*-block ionization energies repeat across periods 2–4.**

These three concerned ionization energies of heavy elements and are resolved by the removals described under R2.1 and R2.3: both source tables are gone from the paper, as is all ionization-energy material. We record the diagnosis rather than leave it unstated, since each was a real defect and the reviewer was right about all three:

- **R2.6** was the two calibrations of R2.1 surfacing — the same atom appearing under the total-energy calibration and the ionization-energy calibration without either being labelled.
- **R2.7** was structural, not a transcription error. With the core absorbed into  $r_c$  and the screened charge identical for every noble-gas valence, each column member presented the *same* valence problem, so the entries repeated by construction: the model reproduced the column and had no representation of the period. That is a serious limitation of the frozen-core construction and is among the reasons we removed it rather than defended it.
- **R2.5** we did not fully diagnose before the construction that produced it was withdrawn.

**R2.8 — Grid convergence is demonstrated only for the helium total energy.**

The reviewer is right that the ladder covered one atom. We have looked beyond it, and the answer is not more rungs. Refining the radial mesh converges the calculation to the answer of the *spherically*

*reduced* problem, and that reduction — not the discretisation — is what limits the accuracy here. The revision measures the reduction's cost directly: it has the pair *count* right and the pair *geometry* wrong, overestimating intra-shell repulsion by a factor of 1.6–2, leaving a residual of order a per cent in the under-binding direction. Against that, successive  $\sqrt{2}$  refinements of helium move the total energy by about 0.2% a step. The model error is an order of magnitude the larger, so a finer radial grid buys precision in solving the wrong equation. The accuracy-limiting direction is *angular* resolution, restoring the pair geometry the reduction discards, and the paper says so instead of offering a mesh ladder as the open question.  $N = 400$  is also what the three-dimensional solver reaches, at  $400^3$ .

**What the refined runs cost us, reported rather than left to be found.** Argon agrees to +0.1% at  $N = 400$ , the closest entry in the table, and it does not survive: on the same domain at  $N = 724$  and  $N = 1024$  it moves to  $-511.7$  and  $-511.3$  Ha, about 3% *under-bound*, the two finest runs agreeing to 0.08% and both step-converged. Neon is comparable in size and non-monotone over the same range. Two changes follow. The paper now states that the table is to be read at the level of a few per cent, with no single entry claimed to better than a per cent; and we have **removed the claim that the model over-binds the heavier third-period atoms with the opposite sign of error to Hartree–Fock**, since the sign of argon's error is a function of the grid. What survives refinement, and what the reduction predicts, is under-binding — the direction of all eighteen fourth-row entries.

## Summary of changes

### Removed

- All ionization-energy material, including Eq. (1)'s Koopmans identification, and the tables underlying R2.5–R2.8.
- The frozen core, the pseudo-kernel,  $r_c$ , and the two-scale core/valence decomposition.
- The "4s before 3d" claim (R2.4) and the "in-principle parameter-free" claim as worded (R2.1), the latter now restated and true.
- The claim that the model over-binds the heavier third-period atoms with the opposite sign of error to Hartree–Fock (R2.8): argon's error changes sign between  $N = 400$  and  $N = 1024$ .
- The sentence Reviewer 2 quoted on the correctness of the radial description.

### Added

- §*Two numbers, not four: the pair and the octet* — the capacities built from the antipodal pair and the 4+4 octet, with the closure argument, the 4+3 step, the noble-gas statement, the direction on the doubling, and Linnett credited. Closure is stated as exhaustion of a row's capacity and not as a completed outer octet, which is false from the fourth row on: krypton's outermost eight is not 4+4 under either filling rule, nor in the orbital account.
- §*Aufbau: how the shell structure forms* — the build-up as what makes energy minimisation select correctly, with the monotonicity measurement and the seed ladder (R2.2).
- A fourth-row table,  $Z = 19\text{--}36$  at  $N = 800$ : mean |error| 3.3%, all eighteen under-bound.
- Table 1 recomputed with no per-element constant: mean |error| 1.6%, maximum 4.5%.
- §*The shape of the computed field: constant charge per unit radius* — the one scaling the construction does not contain: the charge per unit radius is the same in every shell and equal to  $Z/2$ , which fixes the atomic size near 2 a.u. independently of  $Z$  and so returns the near-constancy of atomic radii as a consequence rather than an input.

- A measurement bearing on the doubling: the geometric capacity  $4\pi(r/d)^2$  per shell, on four atoms and three candidate fillings. The pair-and-octet filling is the more self-consistent, but on three informative shells only, so it is recorded as an anticipation of the filling at larger  $Z$  and not as a derivation of the doubling, which remains the principal open item.
- §*Electronegativity from the shell energies* — the Allen configuration energy, replacing ionization as the valence observable (R2.3).
- An analysis of what the spherical reduction costs, with the intra-shell repulsion table (R2.8).
- Three figures of the three-dimensional build-up, none in the original.
- The speculation caveat on ontological claims (R1.1), the molecular limits (R1.3), the basis-set concession (R1.4), and the imaginary-time clarification (R1.5).

## Restated

- Scope in its two senses: the *computed* energies cover periods 1–4, hydrogen to krypton, all-electron and parameter-free; the *account of the build-up* covers every row length and every closed shell. What is bounded at argon is the outward build-up, not the calculation.
- The shell structure as an output of the build-up, not of energy minimisation only.
- The Aufbau result as established for Li–C and open from nitrogen.
- The title, extended to name the new content: *A 3D Multiphase Continuum Model for Atoms and the Periodic Table*. The model is three-dimensional; the periodic-table results are computed with the spherically homogenised reduction of it, as the text states throughout.

The revision retracts what it could not defend and we have tried to save none of it. What replaces it is the construction: the table assembled from an antipodal pair and a dual-tetrahedral octet, every row length and every closure following from what those two units can and cannot compose. That is a different kind of claim from the original's — checkable by composition rather than weighed against a fitted number — and we would ask that the manuscript be judged on it.